STATS 7022 - Data Science PG
Assignment 1

Trimester 1, 2025

Question 1: Data Analysis

A data scientist is often required to quickly produce short analyses. In this question, you will load a dataset and produce a simple analysis by recreating a plot.

Question

The `mpg`[1] dataset contains detailed information about popular car models released between 1999 and 2008, including measures of fuel economy for driving on city roads and highways. Load the `mpg` dataset in to `R`, then write code to exactly reproduce the plot given in Figure 1.

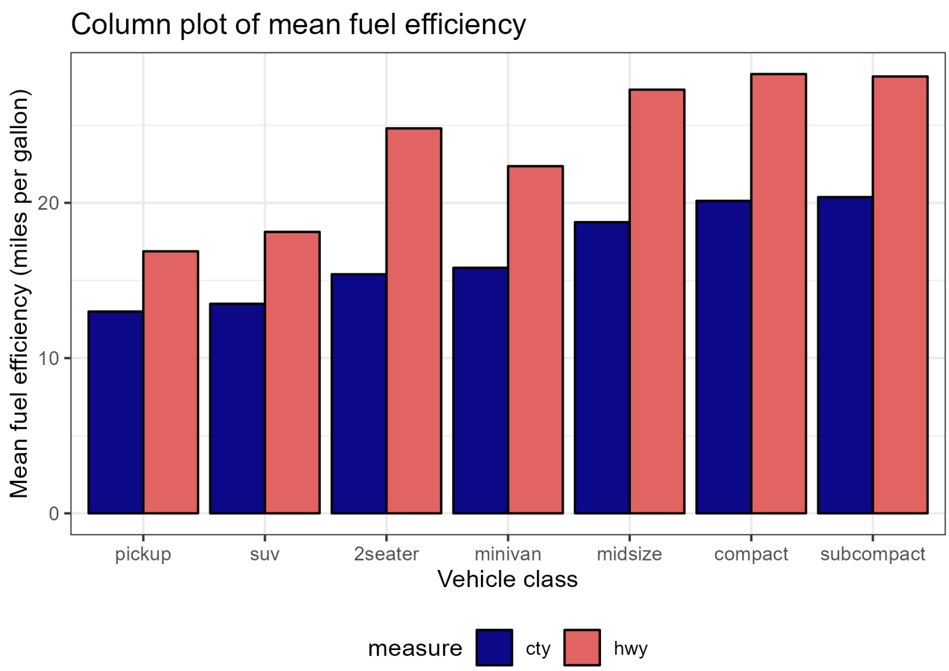

Figure 1: Plot for Assignment 1 Question 1.

For further information, see the assessment rubric on MyUni.

[Question total: 10 marks]

[1]You may load this data in `R` by using the command `data(mpg, package = ‘‘ggplot2’’)`.

Submission

A single pdf file containing both your plot and the `R` code you wrote to generate it.

Checklist

Please ensure that:

- Your submission includes all `R` output and plots to support your answers where necessary.
- Your submission includes all `R` code.
- Your submission includes a caption for every plot and table.
- Your submission is a single pdf file - correctly oriented, clear, and legible.
- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

The course's regular late policy applies to this assignment. If you need to request an extension, or request a variation to the assessment on medical or compassionate grounds, please contact the course coordinator.

Question 2: ROC Curve Derivation

An important part of being a data scientist is understanding the mathematics that underpins the methods we use. In this question, we will look at the theory behind ROC curves.

Question

Consider an analysis with N subjects. The measured outcome is a categorical variable with two levels, A and B. Assume that the proportion of subjects with level A is p.

Rather than fitting a predictive model, suppose you decide to toss a coin and implement the following scheme:

- Toss the coin once per subject.
- If the outcome of the coin toss for subject i is "heads", classify that subject as A.
- If the outcome is "tails", classify that subject as B.

(a) Consider tossing a fair coin. When the fair coin is tossed, the outcome is a head with probability $1/2$. Using the fair coin, calculate:

  (i) the expected sensitivity of your scheme.

  (ii) the expected specificity of your scheme.

(b) Now consider tossing a biased coin. When the biased coin is tossed, the outcome is a head with probability q, $0 \leq q \leq 1$. Recalculate the expected sensitivity and specificity if your scheme uses the biased coin.

(c) From these results, what can you conclude about ROC curves?

[Question total: 10 marks]

Submission

A single pdf file containing your solutions. You may handwrite your answers and scan them, or typeset your answers.

Checklist

Please ensure that:

- Your submission is a single pdf file - correctly oriented, clear, and legible.
- You have shown all of your working, including probability notation where necessary.

- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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Question 3: ROC Function

One of the best ways to understand an algorithm is to implement it yourself. In this question, we will implement the ROC algorithm.

Question

In `R`, write a function called `get_ROC()`. Your function should take two vector inputs:

- `obs`: a vector of the true observed values, `‘‘A’’` and `‘‘B’’`.
- `A`: a vector of predicted probabilities that each observation is `‘‘A’’`.

Your function should treat `‘‘A’’` as positive/success and `‘‘B’’` as negative/failure.

Your function should return a tibble with three columns:

- `threshold`
- `specificity`
- `sensitivity`

Your results in these columns should be consistent with the output of the function `yardstick::roc_curve()`.

Here is an example to illustrate its use:

```
df <- tibble(
  obs = rep(factor(c("A","B")), each = 2),
  A = rep(c(0.8,0.2), each = 2)
)
get_ROC(df$obs, df$A)
```

```
## # A tibble 4 x 3
##   threshold specificity sensitivity
##       <dbl>       <dbl>       <dbl>
## 1    -Inf             0           1
## 2       0.2           0           1
## 3       0.8           1           1
## 4     Inf             1           0
```

Here is a template you can use:

```
# Function to return ROC
#
# INPUT
# Two vectors:
```

```
#   obs: a factor with two levels, "A" and "B"
#   A: the predicted probability that each observation
#      is "A"
#
# OUTPUT
# A tibble with 3 columns: threshold, specificity, and
# sensitivity.

get_ROC <- function(obs, A) {
  # YOUR CODE HERE
}
```

Your function should also ensure that the correct type of input is passed to the function. For further information, see the assessment rubric on MyUni.

[Question total: 10 marks]

Submission

A single R file containing your function. Your code will be automatically unit-tested, so any changes to names etc. may result in a loss of marks. Your function must not use ROC functions from other packages - the code must be your own work.

Checklist

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Question 4: Technical Communication

There are four key skills in a data scientist's arsenal: analysis, mathematics, coding, and communication. In this question, we'll practice your communication skills.

Information can often be effectively communicated with a graph, plot or figure. An effective plot should be easy to read (appropriate use of colours, shapes, and scale) and clearly labelled. It must also be suitable for the types of variables (quantitative or categorical) being considered.

Question

Suppose you are a professional data scientist who has been asked to write an overview of ROC curves for a colleague who has a university degree but only a high-school level of understanding of mathematics, statistics, and programming. Your overview should:

- describe how a ROC curve is calculated;
- explain how to interpret a ROC curve; and
- discuss the advantages and disadvantages of ROC curves.

Marks will be allocated for the quality and correctness of your writing, as well as how suitable your writing is for the intended audience. Your overview must use Harvard-style referencing as detailed here:
https://www.adelaide.edu.au/library/referencing-support

For further information, see the assessment rubric on MyUni.

[Question total: 11 marks]

Submission

A single doc or pdf file containing your overview. Your overview should be 1-2 A4 pages, with 10-12pt font, single line spacing, and standard page margins.

Checklist

Please ensure that:

- Your submission is a single doc or pdf file - correctly oriented, clear, and legible.
- Your submission is not longer than 2 pages (including tables and figures).
- Your submission is correctly referenced. For more information, see the link above or visit the Writing Centre.

- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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Question 5: Variance-Bias Trade-Off

A good data scientist understands the mathematics that underpins the methods they use. In this question, we will derive the variance-bias trade-off.

Good data scientists are also good communicators. This includes communicating mathematical theory. Your answer to this question should use mathematical notation that is clear, consistent, and correct, and you should justify all steps beyond simple algebraic manipulation. Marks will be allocated for the clarity of your answer.

Question

Consider a quantitative response variable Y, and p different predictors $\boldsymbol{x} = (x_1, x_2, \ldots, x_p)$. Assume that the relationship between Y and $\boldsymbol{x}$ can be written in the general form

$$Y = f(\boldsymbol{x}) + \epsilon,$$

where f is some fixed but unknown function of $\boldsymbol{x}$ and ϵ is a random error term that is independent of $\boldsymbol{x}$ and has mean zero. In this formulation, f represents the systematic information that $\boldsymbol{x}$ provides about Y.

Suppose, using some training data, we estimate f as $\hat{f}$. If the testing data comprises a single new observation $(\boldsymbol{x}_0, Y_0)$, independent of the training data, show that the test MSE

$$E\left[(Y_0 - \hat{f}(\boldsymbol{x}_0))^2\right] = \text{var}(\hat{f}(\boldsymbol{x}_0)) + b_f(\hat{f}(\boldsymbol{x}_0))^2 + \text{var}(\epsilon).$$

[Question total: 11 marks]

Submission

A single pdf file containing your solutions. You may handwrite your answers and scan them, or typeset your answers.

Checklist

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